ROLL NO: _____

MADHAV INSTITUTE OF TECHNOLOGY & SCIENCE, GWALIOR

(A Deemed to be University | Autonomous Institute Established in 1957 | NAAC 'A++' Accredited)

MID-SEM EXAMINATION â€" DECEMBER / JUNE 2024

B.TECH. CSE ENGINEERING â€" SEMESTER 4

Time Allowed: 1.5 Hours **Maximum Marks: 30**

NOTE: 1. Attempt any FIVE questions selecting at least ONE question from each Unit.
2. All questions carry equal marks. Sub-questions carry marks indicated in brackets [ ].
3. Assume suitable missing data, if any, and state it clearly.
4. Use of scientific non-programmable calculator is permitted. Mobile phones are strictly prohibited.

--- PART A (Compulsory) ---

Q.1 (a) Define Deterministic Finite Automata (DFA) formally with 5-tuple notation. [3 Marks]
(b) Construct a DFA that accepts all binary strings containing '010' as substring over alphabet {0, 1}. [5 Marks]
(c) Convert the given Non-Deterministic Finite Automata (NFA) with epsilon moves to equivalent DFA. [7 Marks]

[OR]

Q.1 (Alt) State and prove Arden's Theorem. Use it to find regular expression for the given state transition diagram. [5 Marks]

--- PART B (Attempt any ONE) ---

Q.2 State and prove the Pumping Lemma for Regular Languages. Prove that L = {0^n 1^n | n >= 1} is not regular. [10 Marks]

[OR]

Q.2 (Alt) Explain the Chomsky Hierarchy of languages. Construct a Pushdown Automata (PDA) for L = {w c w^R | w in {a, b}*}. [10 Marks]

**** END OF QUESTION PAPER â€" MITS GWALIOR AUTONOMOUS EXAMINATION SYSTEM ****